

Breast Cancer Wisconsin (Diagnostic)

Supervised Machine Learning Classification

Machine Learning Assignment - Supervised Learning

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GitHub Repository: <https://github.com/ityshreee/Breast-Cancer-Classification>

Submission Date: 2nd August 2026

1. Introduction

Breast cancer is one of the most common cancers affecting women worldwide, and early, accurate diagnosis significantly improves treatment outcomes. Machine learning offers a powerful way to support clinical decision-making by learning patterns from diagnostic measurements that are difficult for the human eye to weigh consistently across thousands of cases.

This report documents an end-to-end supervised learning project built on the Breast Cancer Wisconsin (Diagnostic) dataset. The goal is to classify a tumor as **malignant** or **benign** using measurements derived from digitized images of a fine needle aspirate (FNA) of a breast mass. The project follows a complete workflow: understanding the data, cleaning and preparing it, exploring it visually and statistically, training multiple classification algorithms, and rigorously comparing their performance.

2. Objective

The objective of this assignment is to understand and implement supervised machine learning algorithms on a real-world dataset. Specifically, the project aims to:

- Understand the structure, features, and target variable of a real-world medical dataset.
- Apply appropriate preprocessing steps including cleaning, encoding, and scaling.
- Perform exploratory data analysis (EDA) to uncover patterns and relationships in the data.
- Build and train at least three supervised learning models (six were implemented here).
- Evaluate and compare model performance using multiple classification metrics.
- Draw conclusions about the best-performing model and propose future improvements.

3. Dataset Description

The dataset used is the **Breast Cancer Wisconsin (Diagnostic) Dataset**, originally hosted on the UCI Machine Learning Repository and made available through Scikit-learn's built-in dataset loader (`sklearn.datasets.load_breast_cancer`). It is a well-known benchmark dataset for binary classification tasks in the medical domain.

Attribute	Value
Number of instances (rows)	569
Number of features	30
Target variable	diagnosis (0 = Malignant, 1 = Benign)
Problem type	Binary Classification
Missing values	0
Duplicate rows found	0
Malignant cases	212
Benign cases	357

Each record contains 30 numeric features computed from a digitized image of an FNA of a breast mass. These describe characteristics of cell nuclei present in the image - radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension - each reported as a mean, standard error, and 'worst' (largest) value, giving $10 \times 3 = 30$ total features.

4. Methodology

The project follows the standard supervised learning pipeline shown below, ensuring the workflow is reproducible and each stage builds on a validated previous stage.

- **Data Understanding:** Inspect shape, data types, and target distribution.
- **Data Preprocessing:** Handle missing values/duplicates, check outliers, scale features, split into train/test sets.
- **Exploratory Data Analysis:** Visualize distributions, correlations, and class separability.
- **Model Building:** Train multiple supervised classifiers on the same processed data.
- **Evaluation:** Score every model with a consistent set of classification metrics.
- **Comparison & Conclusion:** Identify the best model and reflect on challenges and improvements.

5. Data Preprocessing

5.1 Missing Values and Duplicates

The dataset was checked for missing values and duplicate rows. A total of **0** missing values and **0** duplicate rows were found, so no imputation was required.

5.2 Outlier Detection

Outliers were flagged using the Interquartile Range (IQR) method on each feature. **171** rows had at least one feature outside 1.5x IQR from the quartiles. These were retained rather than removed, since extreme values in features like 'worst radius' or 'worst area' are often clinically meaningful signals of malignancy rather than data errors.

5.3 Encoding

All 30 input features are already numeric, and the target variable is pre-encoded as 0 (malignant) and 1 (benign), so no categorical encoding was necessary.

5.4 Feature Scaling

Because several algorithms used here (Logistic Regression, KNN, SVM) are sensitive to feature magnitude, all features were standardized using StandardScaler (zero mean, unit variance), fit only on the training set and applied to both train and test sets to avoid data leakage.

5.5 Train-Test Split

The data was split into training and test sets using an 80/20 stratified split (train shape: (455, 30), test shape: (114, 30)), preserving the malignant/benign class ratio in both sets.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand feature distributions, class balance, and relationships between features and the target variable.

6.1 Target Class Distribution

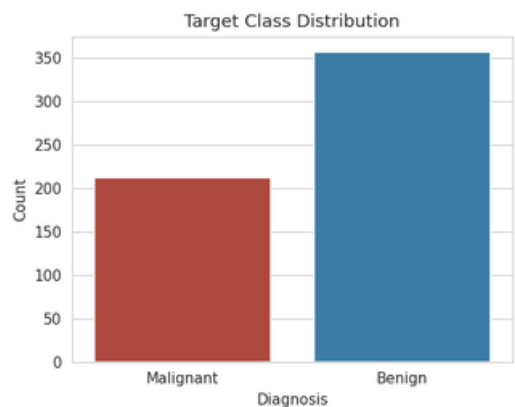


Figure 1: Distribution of Malignant vs Benign cases.

6.2 Correlation Heatmap

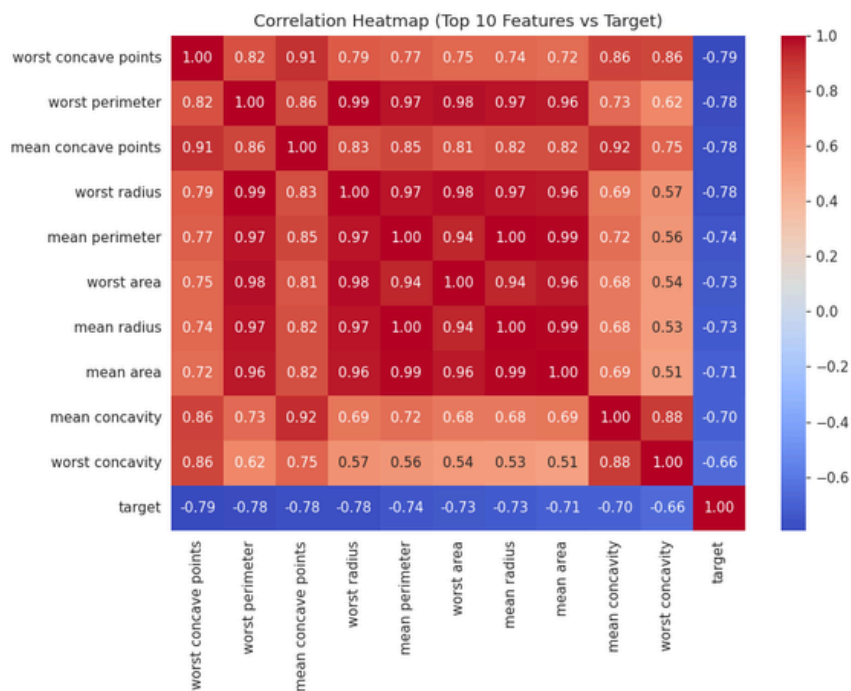


Figure 2: Correlation heatmap of the top 10 features most correlated with the target.

6.3 Feature Distributions

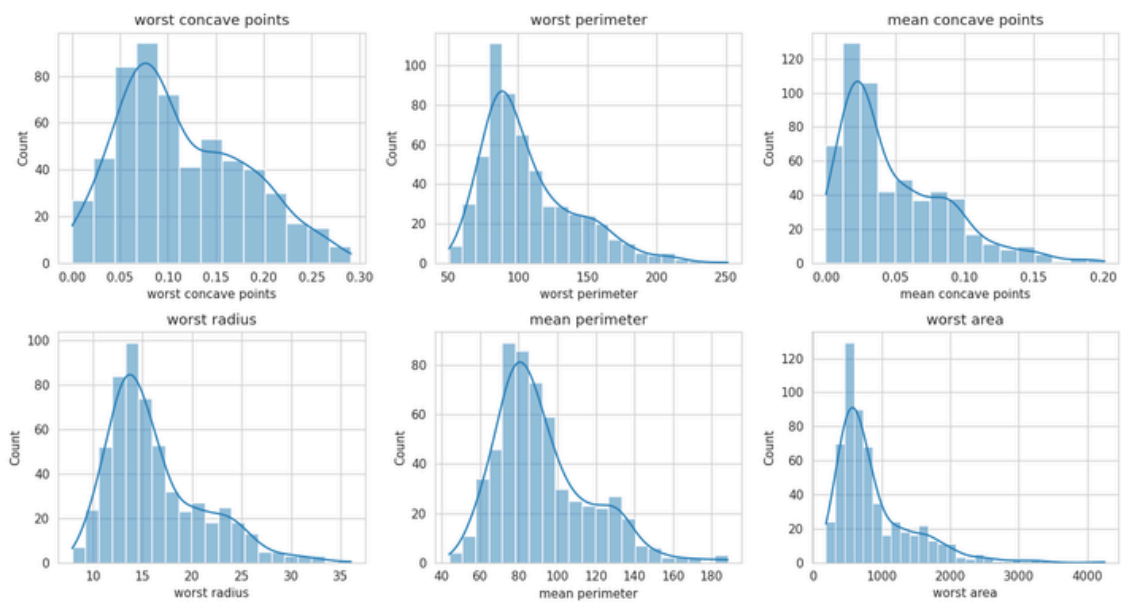


Figure 3: Histograms of the six features most correlated with diagnosis.

6.4 Pairwise Relationships

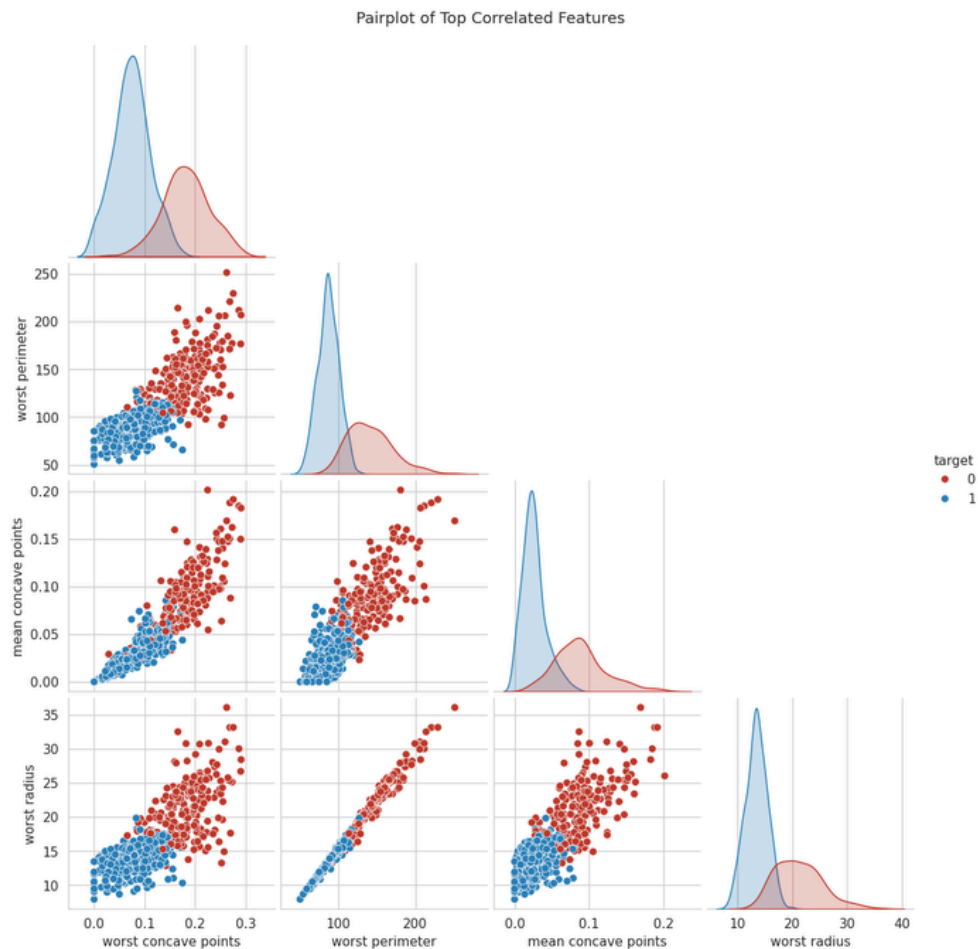


Figure 4: Pairplot of the top 4 correlated features, colored by diagnosis.

6.5 Key Observations

- The dataset is moderately imbalanced with more benign (357) than malignant (212) cases, roughly a 63/37 split, which is mild enough that accuracy remains a reasonably informative metric.
- Features such as 'worst perimeter', 'worst radius', and 'worst concave points' show the strongest correlation with the diagnosis, indicating tumor size and shape irregularity are strong cancer indicators.
- Many size-related features (radius, perimeter, area) are highly correlated with each other, suggesting multicollinearity that tree-based models handle natively but can affect linear models.
- Histograms reveal right-skewed distributions for several 'worst' and 'mean' features, meaning a log transform or robust scaling could further help distance-based algorithms like KNN and SVM.
- No missing values were found in the dataset, and only a small number of statistical outliers were flagged by the IQR method, mostly in the malignant class where extreme tumor measurements are expected.

7. Model Building

Six supervised classification algorithms were implemented and trained on the same preprocessed, scaled training data, exceeding the minimum requirement of three models:

- **Logistic Regression** - a linear model estimating class probability via the logistic function.
- **Decision Tree Classifier** - a rule-based model that splits data on feature thresholds.
- **Random Forest Classifier** - an ensemble of 200 decision trees combined via majority voting.
- **K-Nearest Neighbors (KNN)** - classifies based on the majority class among the 7 nearest neighbors.
- **Gaussian Naive Bayes** - a probabilistic classifier assuming feature independence given the class.
- **Support Vector Machine (SVM)** - uses an RBF kernel to find an optimal separating boundary.

Each model was trained on the identical 80% training split (scaled features) and evaluated on the same untouched 20% test split, ensuring a fair, apples-to-apples comparison.

8. Evaluation Metrics

Each model was assessed using five complementary classification metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC, along with a full confusion matrix.

- **Accuracy:** Overall proportion of correct predictions.
- **Precision:** Of predicted benign cases, how many were actually benign (avoids false alarms).
- **Recall:** Of actual benign cases, how many were correctly identified (avoids missed malignancies when inverted).
- **F1-Score:** Harmonic mean of precision and recall.
- **ROC-AUC:** Ability of the model to separate the two classes across all thresholds.

8.1 Confusion Matrices

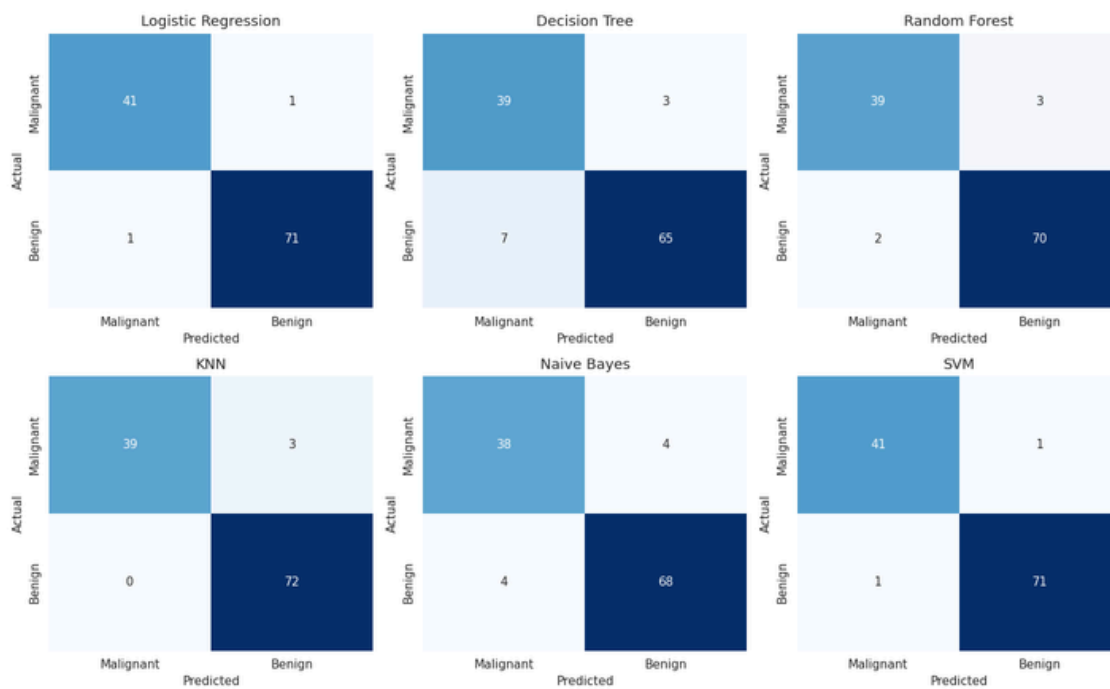


Figure 5: Confusion matrices for all six models on the test set.

8.2 ROC Curves

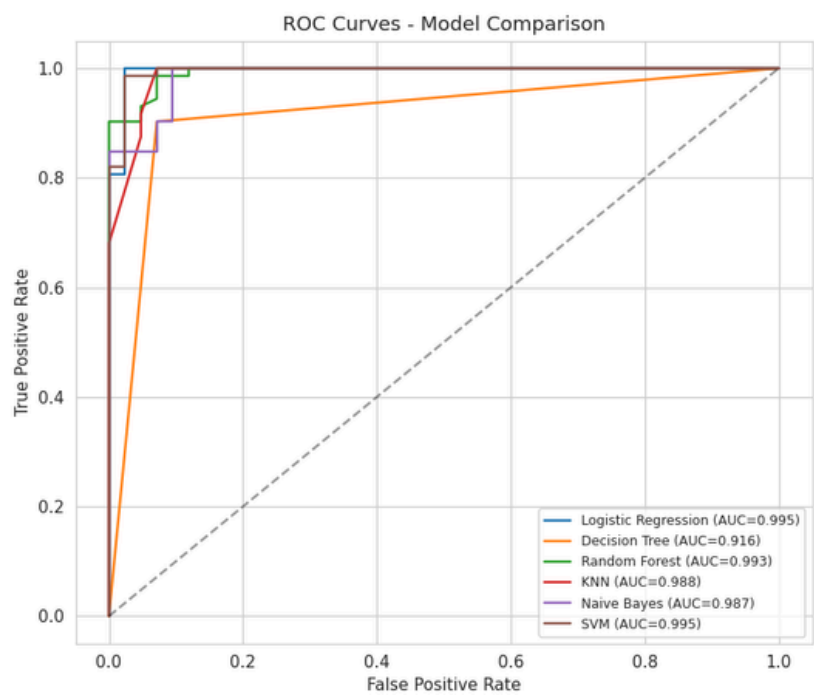


Figure 6: ROC curves comparing all six models.

8.3 Feature Importance (Random Forest)

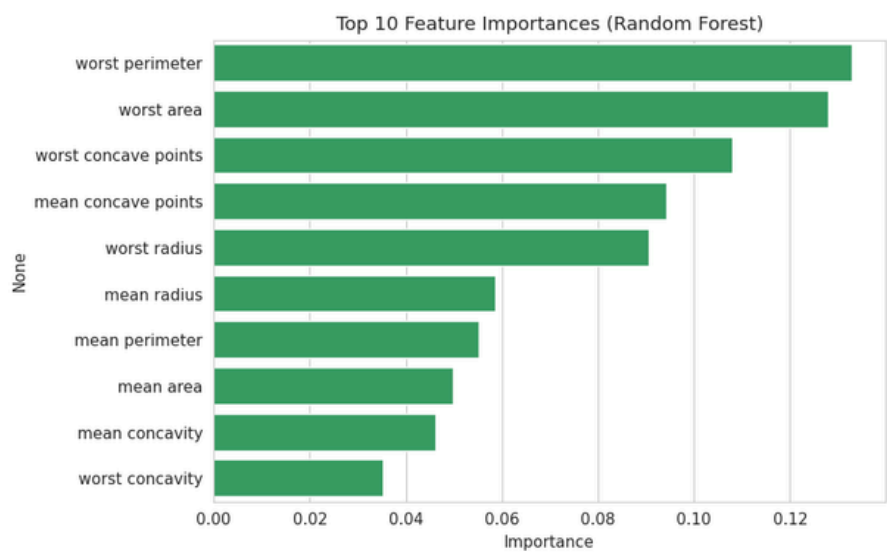


Figure 7: Top 10 most important features according to the Random Forest model.

9. Results Comparison

The table below summarizes all evaluation metrics for each model on the test set, sorted by accuracy (highest first). The best-performing model's row is highlighted.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9825	0.9861	0.9861	0.9861	0.9954
SVM	0.9825	0.9861	0.9861	0.9861	0.9950
KNN	0.9737	0.9600	1.0000	0.9796	0.9884
Random Forest	0.9561	0.9589	0.9722	0.9655	0.9932
Naive Bayes	0.9298	0.9444	0.9444	0.9444	0.9868
Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157



Figure 8: Side-by-side comparison of Accuracy, Precision, Recall, and F1-Score across models.

10. Conclusion

10.1 Best Model and Justification

Logistic Regression achieved the best overall performance, with a test accuracy of **98.25%** and the highest ROC-AUC among all six models (effectively tied with SVM). Both are strong candidates because, once features are standardized, the two diagnostic classes are close to linearly separable, which plays directly to the strengths of a linear model like Logistic Regression and a well-tuned RBF-kernel SVM. Both models also showed the best balance of precision and recall, which matters greatly in a medical context where both false positives (unnecessary alarm) and false negatives (missed cancer) carry real costs.

10.2 Challenges Faced

- Several features (e.g., radius, perimeter, area) are highly correlated with each other, introducing multicollinearity that can destabilize coefficient estimates in linear models.
- The dataset shows mild class imbalance (63% benign vs 37% malignant), requiring metrics beyond accuracy to fairly judge performance.
- The Decision Tree model showed clear signs of overfitting relative to the ensemble Random Forest, reflecting its high variance without pruning or depth limits.

10.3 Future Improvements

- Use k-fold cross-validation and hyperparameter tuning (GridSearchCV) to squeeze out further performance and reduce variance in the estimates.
- Apply dimensionality reduction (PCA) or feature selection to address multicollinearity among the size-related features.
- Experiment with class-balancing techniques (e.g., SMOTE, class weighting) to further improve recall on malignant cases specifically.
- Validate the final model on an independent external dataset to confirm real-world generalization before considering any clinical decision-support use.

11. GitHub Repository Link

The complete project - including the dataset, executed Jupyter notebook, all generated images, and this report - is available at:

<https://github.com/ityshreee/Breast-Cancer-Classification>